INCH-POUND

MIL-C-28731D
AMENDMENT 2
11 October 1991
SUPERSEDING
AMENDMENT 1
2 MAY 1986

MILITARY SPECIFICATION

CONNECTORS, ELECTRICAL, RECTANGULAR,
REMOVABLE CONTACT, FORMED BLADE, FORK TYPE
(FOR RACK AND PANEL AND OTHER APPLICATIONS)
GENERAL SPECIFICATION FOR

This amendment forms part of MIL-C-28731D, dated 4 June 1985, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, following MIL-C-22529/17, add:

"MIL-M-24519 - Molding plastics, Electrical, Thermoplastic."

PAGE 2

2.1, following FED-STD-66, add:

"O-T-634- Trichloroethylene Technical."

PAGE 3

3.3.5, delete and substitute:

"3.3.5 Insert material. Unless otherwise specified (see 3.1), insert material shall be a diallyl phthalate in accordance with MIL-M-14, type SDG-F, GDI-30F, or polyester thermoplastic conforming to MIL-M-24519. (Reground material shall not be used.)"

PAGE 7

After 3.21, add new paragraphs 3.22 and 3.23 as follows:

"3.22 Resistance to soldering heat. Connectors shall withstand the tests specified in 4.7.16.

"3.23 Resistance to solvents. Connectors shall withstand the tests specified in 4.7.17."

Table III, Applicable specification column, Plastic: Add "MIL-M-24519".

* After 4.1.1, add new paragraph as follows:

"4.1.2 Assembly plants. Assembly plants must be listed on or approved for listing on the applicable Qualified Parts List. The qualified connector manufacturer shall certify that the assembly plant is approved for the distribution of the manufacturer's parts. The assembly plant shall use only piece parts supplied by the qualified connector manufacturer. No testing other than visual examination is required of certified piece parts obtained from the qualified manufacturer, except when there is cause for rejection. All assemblies produced at the assembly plant shall be subjected to examination of product to assure that the assembly process conforms with that established at the qualified manufacturing plant. Quality control requirements, including Government inspection surveillance, shall be the same as required for the qualified connector manufacturer."

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TABLE IV, delete and substitute:

" TABLE IV. Qualification inspection - Connectors.

Inspection	Requirement paragraph	Test method paragraph	Groups 1/		Conditions
			1	2	
Visual and mechanical inspection	3.1, 3.3, 3.4, through 3.4.5, 3.5, 3.20, and 3.2	4.7.1	x	x	All connectors
Insulation resistance	3.6	4.7.2	x	------	Unmated
Temperature cycling	3.7	4.7.3	x	------	Mated
Dielectric withstanding voltage	3.8	4.7.4	x	------	Unmated
Contact retention	3.9	4.7.5	x	------	Unmated
Vibration	3.10	4.7.6	x	------	Unmated
Mechanical shock	3.11	4.7.7	------	x	Mated
Mating and unmating forces	3.12	4.7.8	------	x	Mated or unmated
Humidity	3.13	4.7.9	x	------	Unmated
Durability	3.14	4.7.10	x	------	Mated or unmated
Salt spray	3.15	4.7.11	x	------	Mated
Contact resistance	3.16	4.7.12	x	------	Mated
Resistance to soldering heat 2/	3.22	4.7.16	------	x	Unmated
Resistance to solvents	3.23	4.7.17	------	x	Unmated

1/ For group qualification, three sample of the largest size and one sample of each smaller size hall be subjected to group and 2 tests.
2/ For solder type terminations only."

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After 4.7.15, add new paragraphs 4.7.16 and 4.7.17 as follows:

"4.7.16 Resistance to soldering heat (see 3.22). All connectors with solder terminations shall be tested in accordance with method 210 of MIL-STD-202, condition C.

a. The applicable wire size properly prepared shall be inserted into the contact termination. A minimum of seven contacts shall be tested.

b. An appropriately prepared resistance soldering iron with an appropriate tip shall be applied to the lower portion of the terminal configuration where the wire enters the termination.

c. The solder shall be applied in the normal manner.

d. The resistance soldering iron shall be applied to the system. The wattage shall be adjusted as to allow a proper solder fillet to be formed, or for a 4-second minimum time limit, whichever is more.

e. After application, the soldering iron shall be removed and a visual and mechanical inspection preformed.

f. Visual inspection shall be at 1X magnification. The connector shall show no evidence of distortion or damage to any area of the connector housing. The contact shall meet the contact retention requirements (4.7.5). The contact shall not interface with normal floating conditions as applicable and shall meet applicable location dimensions.

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4.7.17 Resistance to solvents (see 3.23). Connectors shall be tested in accordance with the following details.

a. An appropriate size beaker large enough to accept the connectors to be tested shall be elected.

b. The beaker shall be filled to half capacity or to a level which shall completely cover the connectors, which is less, with trichloroethylene per O-T-634, type II.

c. The solution shall be heated to its boiling point and shall be so heated for a 15-minute time period prior to exposing the connectors and shall be maintained at that level for the duration of the test.

d. The connectors shall be placed in a wire mesh basket of an appropriate size.

e. The wire mesh basket shall be subjected to the following schedule:

1. Expose to fumes for 18 seconds.

2. Immediately lowered into the solution for 42 seconds.

3. Reexposed to fumes for 18 seconds.

4. Removed from test beaker.

f. The connectors shall be allowed to cool to room temperature.

g. The connectors shall be examined under 1X magnification. There shall be no evidence of cracking, crazing, discoloration, distortion, or bleeding out of any foreign matter from the material. Pitting shall not be allowed. Change of the surface from bright to a matte finish shall not be construed as a failure.

h. Connectors so tested shall be fully assembled by normal assembly operations."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notions. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air force - 85

Review activities:
Army _ AR, AT, MI
Navy - AS, OS, SH
Air Force - 11, 99
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5935-3846)

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